

Chapter

Soil Resources

1. Which of the following helps in soil conservation? MAIN 2024

- (a) Afforestation
- (b) Overgrazing
- (c) Mining
- (d) Shifting agriculture

2. Which is the most widespread soil of India? MAIN 2023

- (a) Red soil
- (b) Alluvial soil
- (c) Laterite soil
- (d) Black soil

3. Name the soil that is found in central Deccan plateau. SQP 2023

- (a) Black soil
- (b) Alluvial soil
- (c) Red soil
- (d) Laterite soil

4. How is the soil named by you in Q. 23 above formed? COMP 2021

- (a) Formed by the deposition of silt brought down by rivers
- (b) Formed by the weathering of laterite rocks.
- (c) Formed by the weathering of basic lava rocks
- (d) Formed by the weathering of metamorphic rocks

5. Why are the foothills of Himalayas prone to excessive soil erosion ?

- (a) Absence of crop cover
- (b) To much of development of transportation

- (c) Large scale deforestation in past
- (d) Tourists are spreading pollutants in rivers

MAIN 2021

6. Name the soil that is generally confined to river basins. MAIN 2021

- (a) Black soil
- (b) Alluvial soil
- (c) Red soil
- (d) Laterite soil

7. Name the soil that becomes infertile because of desilication. SQP 2021

- (a) Black soil
- (b) Alluvial soil
- (c) Red soil
- (d) Laterite soil

8. A type of leached soil is: COMP 2022

- (a) Red
- (b) Black
- (c) Alluvial
- (d) Laterite

9. Why is the Red soil considered ideal for dry farming? SQP 2023

- (a) It is formed in areas with poor rainfall and responds to fertilizers.
- (b) It is red in colour.
- (c) It is rich in iron oxide.
- (d) It is rich in nitrogen.

**10. With reference to Laterite soil, answer the questions that follow:
MAIN 2024**

- (i) How is Laterite soil formed?
- (ii) How can Laterite soil be made productive?
- (iii) Name one region where Laterite soil is found.

11. (i) Mention two causes of soil erosion in India.

(ii) Mention a soil conservation method suitable for hilly area. MAIN 2024

12. Name the soil : MAIN 2023

- (i) That is found on the summits of Eastern and Western Ghats.
- (ii) That is suitable for dry farming.
- (iii) That varies in texture.

13. Give a geographical reason for each of the following: MAIN 2023

- (i) Terrace farming is an ideal soil conservation method for hilly areas.
- (ii) Khadar soils are preferred over Bhangar soils.
- (iii) Foothills of Himalayas have been affected by soil erosion

14. Give one difference between each of the following: MAIN 2024

- (i) Alluvial soil of Uttarakhand and alluvial soil of West Bengal.
- (ii) transported soil and residual soil.

15. How is red soil formed? State two reasons for the low productivity of red soil. COMP 2021

16. How is alluvial soil formed? Why is this soil agriculturally important? SQP 2024

17. Name one soil of volcanic origin commonly found in India. Name one crop widely grown on this soil. COMP 2023

18. What is the soil ? Name the four types of soil. MAIN 2023

19. How are alluvial soil formed ? MAIN 2022

20. What are the characteristics of alluvial soil ? MAIN 2024

21. Mention any two causes of soil erosion in India. MAIN 2023

22. Name two soil conservation methods that may be practised in arid (dry) regions. MAIN 2023

23. (i) How does deforestation lead to soil erosion ? MAIN 2021

- (ii) State two methods of controlling erosion of soil caused by running water.
- (iii) Name two important agents of erosion. For each, state one method of controlling the erosion caused.

24. What are the causes of soil erosion in : SQP 2022

(i) Shiwaliks or the Outer Himalayas.

(ii) North-Eastern parts of India.

(iii) Arid regions of India.

(iv) Hilly areas of the South.

25. (i) Name the soil which occupies largest area after the alluvial soil.

(ii) What is dry farming ?

(iii) Why is the desert soil alkaline in nature? o 'the desert soil contains high proportion of salts'. Why ?

COMP 2023

26. Study the picture given and then answer the follow questions.

(i) What is the type of farming shown above?

(ii) How is the type of farming helpful to agriculture?

(iii) Name another method used which is similar to the one shown above. MAIN 2024



Solution

1. (a) Afforestation

Afforestation helps in soil conservation.

2. (b) Alluvial soil

The most widespread soil of India is **Alluvial soil**.

3. (a) Black soil

The soil found in the central Deccan Plateau is **Black soil**.

4. (c) Formed by the weathering of basic lava rocks

The black soil in the central Deccan Plateau is **formed by the weathering of basic lava rocks**.

5. (c) Large scale deforestation in past

The foothills of the Himalayas are prone to excessive soil erosion due to **large-scale deforestation in the past**.

6. (b) Alluvial soil

The soil that is generally confined to river basins is **Alluvial soil**.

7. (d) Laterite soil

The soil that becomes infertile because of desilication is **Laterite soil**.

8. (d) Laterite

A type of leached soil is **Laterite soil**.

9. (a) It is formed in areas with poor rainfall and responds to fertilizers.

Red soil is considered ideal for dry farming because **it is formed in areas with poor rainfall and responds well to fertilizers**.

10. (i) Laterite soil is formed through **intense leaching** in areas with heavy rainfall and high temperatures, which removes silica and accumulates **iron and aluminum oxides**.

(ii) Laterite soil can be made productive by **adding fertilizers and organic manure** and using **proper irrigation techniques** to improve fertility.

(iii) One region where Laterite soil is found is **Tamil Nadu**.

11. (i) Two causes of soil erosion in India are:

(a) **Deforestation**: Removal of trees reduces protective cover, making soil prone to erosion by wind and water.

(b) **Overgrazing**: Excessive grazing removes vegetation, exposing soil to erosion.

(ii) A soil conservation method suitable for hilly areas is **terrace farming**, which creates step-like terraces on slopes to reduce water runoff and conserve soil.

12. (i) **Laterite soil** – Found on the summits of the Eastern and Western Ghats.

(ii) **Red soil** – Suitable for dry farming due to its moisture-retentive properties.

(iii) **Alluvial soil** – Varies in texture, ranging from sandy loam to clay.

13. (i) Terrace farming is ideal for hilly areas because it **slows down the flow of water** on slopes, reducing runoff and **preventing soil erosion** by allowing more water to soak into the soil.

(ii) Khadar soils are preferred over Bhangar soils because **Khadar is newer, more fertile, and enriched every year by fresh alluvium**, while **Bhangar is older and less fertile** with more coarse materials.

(iii) The foothills of the Himalayas face soil erosion due to **heavy rainfall, deforestation, and overgrazing**, which loosen the soil and make it easily washed away.

14. (i) **Alluvial soil of Uttarakhand** is **coarser in texture** (because it is found in the foothills and carries larger sediments).

Alluvial soil of West Bengal is **finer in texture** (because it is deposited in the lower plains by slow-moving rivers).

(ii) **Transported soil** is the soil **carried from its place of formation** by agents like wind, water, or glaciers (**ex-situ**).

Residual soil is the soil **formed and found at the same place** where the parent rock has weathered (**in-situ**).

15. Here is your answer in a clear, correct, ICSE-ready format:

How is red soil formed?

Red soil is formed by the **weathering of ancient crystalline and metamorphic rocks**, especially **granite and gneiss**, under **hot and dry climatic conditions**. Its red colour is due to the **high presence of iron oxide**.

Two reasons for the low productivity of red soil:

1. **Nutrient deficiency:**

Red soil is **poor in essential nutrients** like nitrogen, phosphorus, humus, and organic matter, which reduces its fertility.

2. **Low moisture retention:**

Because red soil is **porous and sandy**, it cannot retain water well, making it unsuitable for crops that require more moisture.

✓ Perfect for ICSE Geography answers.

16. Alluvial soil forms from the deposition of silt carried by rivers from the mountains. It is agriculturally significant due to its softness, fertility, and richness in humus and lime.

17. **Name one soil of volcanic origin commonly found in India. Name one crop widely grown on this soil.** **Black soil** (also called *Regur soil*) is of volcanic origin and is commonly found in India.

A major crop grown on this soil is **cotton**.

18. **What is soil? Name the four types of soil.**

Soil is a thin, loose layer of rock particles mixed with decomposed organic matter (humus) that covers the Earth's surface. It supports plant growth and is essential for all living organisms.

The four main types of soil found in India are:

- (i) **Alluvial soil**
- (ii) **Black soil**
- (iii) **Red soil**
- (iv) **Laterite soil**

19. Alluvial soil is formed from fine silt, sand, and clay brought down from the mountains by rivers and their tributaries. As the rivers overflow or slow down, they deposit these materials on the plains, forming fertile alluvial soil.

20. (i) Formation & Depth:

Alluvial soil is formed by the deposition of silt, sand, and clay carried by rivers from the upper to the lower course. It is mainly found in the northern plains and piedmont regions and can be as deep as **500 metres**.

(ii) Texture Variation:

In the upper river valleys (near the foothills), the soil is **coarse and sandy**, while in the lower plains it becomes **finer, light, porous**, and **easy to cultivate**.

(iii) Fertility:

Alluvial soil is **highly fertile**, rich in potash, lime, and phosphoric acid, but usually **low in nitrogen and humus**. The **Ganga Delta**, however, has humus-rich alluvial soil.

(iv) Agricultural Importance:

It is suitable for a large variety of crops—both **rabi and kharif**—although in drier regions the soil may become **slightly alkaline**.

21. Two major causes of soil erosion in India are:

- (i) Deforestation
- (ii) Overgrazing

22. Planting of shelter belts and Restrictions on overgrazing of animals.

23. (i) How does deforestation lead to soil erosion?

Deforestation removes trees and vegetation that hold soil together. Without roots to bind the soil, it becomes loose and is easily **washed away by rain or blown away by wind**, leading to soil erosion.

(ii) **Two methods of controlling erosion caused by running water:**

1. **Terrace farming:** Creating step-like terraces on slopes reduces the speed of water flow and prevents soil erosion.
2. **Contour ploughing:** Ploughing along the natural contours of the land slows down water runoff and reduces erosion.

(iii) **Two important agents of erosion and control methods:**

1. **Water:** Causes erosion through rivers, rainfall, and floods. **Control method:** Build **check dams, bunds, and terraces** to slow water flow.
2. **Wind:** Causes erosion in dry and sandy areas by blowing away loose soil. **Control method:** Plant **windbreaks** (trees or shrubs) to reduce wind speed and protect soil.

24. Causes of soil erosion in different regions of India:

(i) **Shiwaliks or the Outer Himalayas:**

Destruction of vegetation – removal of trees and plants loosens the soil, making it easily washed away by running water.

(ii) **North-Eastern parts of India:**

Heavy rainfall and flooding – intense and frequent rainfall leads to soil being carried away by floods and runoff.

(iii) **Arid regions of India:**

Wind erosion – in deserts and dry areas with little vegetation, wind blows away fine sand and soil particles.

(iv) **Hilly areas of the South:**

Steep slopes and poor farming practices – steep terrain, heavy rains, and unscientific agriculture contribute to soil erosion.

25. (i) The soil which occupies the largest area after alluvial soil is **Black soil (Regur soil).**

(ii) **Dry farming** is a method of farming in regions with limited moisture, where crops are grown using **soil moisture conservation techniques** and **drought-resistant crops**.

(iii) Desert soil is **alkaline** in nature because it contains a **high proportion of soluble salts**. High evaporation and low rainfall prevent the leaching of these salts, leading to their accumulation.

26. (i) terrace farming

(ii) Prevents soil erosion

(iii) Contour Ploughing